

Arun Nathi

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SUMMARY

Data Scientist with 3.5+ years of experience designing and implementing AI and machine learning solutions in financial services. Skilled in Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, LSTM networks, and Hugging Face Transformers. Experienced in developing predictive credit and loan risk models analyzing 8,500+ client portfolios, building fraud detection pipelines processing 150,000+ daily transactions, and implementing RAG-based LLM pipelines to extract insights from 500,000+ financial reports, accelerating decision-making 3× for risk teams. Proficient in NLP-driven financial summarization, customer segmentation, and behavior modeling for personalized recommendations. Expertise in explainable AI (SHAP, LIME) to ensure transparency and compliance, as well as end-to-end ML workflows from feature engineering to model deployment.

SKILLS

Machine Learning & Predictive Modeling: Random Forest, XGBoost, Gradient Boosting, Regression (Linear & Logistic), Decision Trees, SVM, Time Series Forecasting, Model Evaluation, Hyperparameter Optimization

Programming & Development: Python, R, NumPy, Pandas, SciPy, Scikit-learn, TensorFlow, PyTorch, Keras, FastAPI, JupyterLab

Artificial Intelligence & Multimodal Learning: GPT-4, LLaMA, LangChain, CLIP, Whisper, BLIP-2, RAG (Retrieval-Augmented Generation), BERT, Transformer Architectures, Vision-Language Models, Prompt Engineering

Generative AI & NLP: Hugging Face Transformers, Semantic Search, Named Entity Recognition (NER), Text Summarization, Embedding Models, Chatbot Frameworks, Vector Databases (Weaviate, FAISS)

Data Management & Engineering: SQL (PostgreSQL, MySQL), NoSQL (MongoDB), Feature Engineering, Data Warehousing, ETL Development

MLOps & Model Deployment: MLflow, Docker, Kubernetes, REST API Integration, CI/CD (Jenkins, GitHub Actions), Model Serving & Monitoring

Cloud & Data Platforms: AWS (SageMaker, Lambda, EC2, S3), Azure ML, GCP (Vertex AI, BigQuery), Databricks, Snowflake

Data Visualization & Reporting: Power BI, Tableau, Seaborn, Matplotlib, Plotly, Google Data Studio, Advanced Excel

EXPERIENCE

Capital One | AI Data Scientist

Nov 2024 – Present | Texas

- Developed predictive credit risk and default models using Python, XGBoost, Scikit-learn, analyzing 250,000 daily transactions across 12 products, enabling analysts to prioritize high-risk accounts and reduce review cycles from 3 days to 18 hours.
- Engineered fraud detection pipelines with PyTorch LSTM and Gradient Boosting, processing 180,000 transactions per day, identifying anomalous patterns and reducing investigation backlog for operational efficiency.
- Implemented NLP-driven financial insights extraction using spaCy, BERT, Transformers, processing 1.2 million customer support interactions and regulatory documents, providing summarized insights to risk and compliance teams for faster decision-making.
- Built customer segmentation and behavior models using K-Means clustering and PCA, analyzing 550,000 customer profiles, generating clusters for personalized credit offers and recommendations across 8 regional units.
- Implemented RAG-based financial intelligence system using LangChain, retrieving insights from 500,000+ reports and documents, enabling risk teams to generate recommendations 3× faster.
- Applied Hugging Face Transformers for financial text classification and sentiment analysis, extracting insights from 1M+ financial reports and customer interactions, improving automated reporting and monitoring workflows.
- Collaborated with teams to integrate machine learning models into internal APIs using FastAPI and Docker, automating decision support for 25 banking products, reducing credit approval time from 24 hours to 6 hours, enabling operational scalability.

LTIMindtree | Data Scientist

Jan 2021 – Jul 2023 | India

- Built and deployed credit risk models using Random Forest, Logistic Regression, and XGBoost on 12 million+ loan records, enhancing accuracy of credit score classification across multiple lending portfolios.
- Designed a customer segmentation pipeline using K-Means and Gradient Boosting, analyzing 8 million account profiles to support personalized financial product recommendations.
- Trained a loan default prediction model in TensorFlow, leveraging time-series data of 600K borrowers to anticipate repayment trends and reduce delinquency risk.
- Engineered data preprocessing and feature extraction workflows with Python, Pandas, NumPy, and SQL, transforming 5 TB of raw financial and demographic data into machine-learning-ready formats.
- Developed a fraud detection system using PyTorch and LSTM models to process 5 years of transaction logs, identifying anomalies across 3 key digital payment channels and reducing false alarms.
- Implemented NLP-based text mining using BERT and spaCy to extract patterns from over 1.2 million customer complaints and support tickets, helping the business uncover key drivers of dissatisfaction.
- Automated model lifecycle management with MLflow and Docker, integrating models into AWS SageMaker for scheduled retraining on weekly financial data batches to maintain stability and accuracy.
- Created interactive Power BI dashboards visualizing loan performance, customer lifetime value, and churn indicators across 6 regional business units, enabling timely executive decision-making.
- Applied model interpretability frameworks (SHAP, LIME) to evaluate feature importance and ensure audit-readiness for all high-risk predictive models.

ACADEMIC PROJECTS

AI-Driven Financial Risk Intelligence System

- Designed and fine-tuned ML models using XGBoost, Random Forest, and Transformer architectures to forecast credit default risk, portfolio volatility, and market exposure, improving model precision by 0.12 AUC on benchmark datasets.
- Built an LLM-based financial summarization engine using LangChain and GPT-4, analyzing earnings reports and investor briefings to extract sentiment, key metrics, and risk indicators, accelerating analysis speed 4x for financial teams.
- Developed interpretable AI models with SHAP and LIME for explainability, enabling transparent decision-making and compliance with SR 11-7 and Fair Lending Guidelines while maintaining predictive accuracy.

EDUCATION

Master of Science in Data Engineering | University of North Texas, Denton, TX